

Úkol 11.1

$A^T A$ má vl. č. 9 (alg. nás. 2) a 3.

$$M_9 = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\} \quad M_3 = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

ON báze M_9

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

OG projekce v_2 na $\text{Span} \{u_1\}$

$$\vec{w} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$u_2' = \vec{v}_2 - \vec{w} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\|u_2'\| = \frac{\sqrt{6}}{2}$$

$$u_2 = \frac{u_2'}{\|u_2'\|} = \frac{1}{\sqrt{6}} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

M_3

$$u_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(u_1, u_2, u_3) je ON báze $\mathbb{R}^3$

$$U = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$D^T = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix}$$

Označím sloupce $V = (v_1, \dots, v_3)$

$$v_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$$v_2 = \frac{1}{\sigma_2} A^T u_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ 1 \\ -2 \\ 1 \end{pmatrix}$$

$$v_3 = \frac{1}{\sigma_3} A^T u_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$V = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 \\ 0 & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 \end{pmatrix}$$

Nalezli jsme tedy singulární rozklad A^T ,
protože $A^T = V D^T U^T$
ždruven $A = U D V^T$

$$A \cdot A^T = U D V^T V D^T U^T = U \cdot D \cdot D^T U^T$$

$$D^T \cdot D = D \cdot D^T = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$A^T A = V D^T U^T U D V^T = V \cdot D^T \cdot D \cdot V^T$$

(všechny matice jsou reálné, takže hermitovské sdružení znamená transpozici)